

Deep Learning Final Exam Projects

2026-2027

Project ID	Project Name	Project Requirements
1	Label-Efficient Image Classification with Vision Foundation Models	- Investigate how pretrained vision foundation models perform when only a limited amount of labeled data is available. - Compare conventional transfer learning with a pretrained vision foundation model using different amounts of labeled training data.
2	Fine-Grained Image Recognition under Limited Training Data	- Investigate the classification of visually similar categories when only a limited amount of labeled training data is available. - Analyze the effects of transfer learning, data augmentation, and training-data size on classification performance.
3	Satellite Image Classification across Different Domains	- Investigate how image-classification models perform when the training and test images come from different regions, conditions, or data sources. - Compare conventional CNN-based models with pretrained vision models under these changes in data distribution.
4	Industrial Visual Anomaly Detection with Pretrained Features	- Detect abnormal or defective products when most training images contain only normal samples. - Compare conventional anomaly-detection methods with approaches based on pretrained vision models.
5	Robust Image Recognition under Realistic Image Degradations	- Investigate how image-classification models perform when input images are affected by blur, noise, low light, or other common degradations. - Compare the robustness of different deep-learning models under changing image quality.
6	Open-Vocabulary Object Detection with Text Prompts	- Detect objects using text descriptions instead of relying only on a fixed set of predefined classes. - Investigate how different object names, synonyms, and descriptive prompts affect detection performance.
7	Effects of User Prompts on Interactive Image Segmentation	- Investigate how different types and qualities of user prompts affect image-segmentation performance. - Compare point prompts, box prompts, and inaccurate prompts, and analyze common failure cases.
8	Applying Pretrained Vision Models across Different Image Domains	- Investigate how well pretrained vision models can be applied to different types of image data. - Compare model performance when the training and test images come from different domains, such as natural, satellite, medical, or industrial images.
9	Vietnamese Image Captioning with Vision-Language Models	- Generate Vietnamese descriptions for images using lightweight vision-language models.

		- Evaluate caption quality and investigate how prompt design affects the generated descriptions and common errors.
10	Visual Question Answering across Different Question Types	- Develop a system that answers text-based questions about image content. - Evaluate performance across different question types, including object recognition, color, counting, spatial relationships, and visual reasoning.
11	Visual Question Answering for Receipts and Documents	- Extract and understand information from receipt and document images using deep-learning models. - Compare traditional OCR-based methods with modern vision-language approaches for answering questions about document content.
12	Chart Understanding with Vision-Language Models	- Investigate how vision-language models understand and answer questions about bar charts, line charts, and other common charts. - Evaluate performance on factual questions, numerical questions, and visual reasoning, and analyze common failure cases.
13	Multimodal Meme Understanding: Image, Text, or Both?	- Investigate how visual and textual information contribute to understanding the meaning of internet memes. - Compare image-only, text-only, and multimodal models, and analyze when combining both modalities improves performance.
14	Multimodal Food Understanding from Images and Text	- Recognize food categories and extract useful information from both food images and related text. - Compare image-only, text-only, and multimodal approaches to determine whether combining both modalities improves food recognition and understanding.
15	Video Question Answering with Multimodal Models	- Answer text-based questions about short video clips using lightweight pretrained multimodal models. - Investigate how frame selection, temporal information, and different question types affect answering performance.
16	Deep Image Inpainting under Different Missing-Region Patterns	- Restore missing or damaged regions in images using deep-learning models. - Compare restoration performance across different mask shapes, mask sizes, and missing-region patterns.
17	Image Super-Resolution under Unknown Degradations	- Recover high-resolution images from degraded low-resolution inputs using deep-learning models. - Evaluate how super-resolution models perform when test images contain blur, noise, or compression conditions that were not included during training.
18	Low-Light Image Enhancement and Downstream Recognition	- Enhance images captured under low-light conditions using deep-learning models. - Investigate whether improved visual quality also leads to better performance on downstream recognition tasks.

19	Can Synthetic Images Improve Imbalanced Classification?	- Generate synthetic images for minority classes to reduce class imbalance in the training data. - Compare synthetic-data augmentation with conventional data augmentation and oversampling, and evaluate their effects on classification performance.
20	Personalizing Text-to-Image Models with Lightweight Fine-Tuning	- Adapt a pretrained text-to-image model to generate images of a new visual concept using only a small training set. - Investigate how the amount of training data and different LoRA settings affect the quality and consistency of generated images.
21	Deep Restoration of Old and Damaged Photographs	- Restore old photographs affected by noise, scratches, missing regions, or other visual degradations using deep-learning models. - Investigate how different restoration steps or restoration pipelines influence the final visual quality.
22	Can a Neural Network Know When It Does Not Know?	- Investigate whether a neural network can recognize inputs that are different from the data used during training. - Compare different methods for detecting unfamiliar or out-of-distribution inputs based on model confidence and learned features.
23	When Can We Trust Neural Network Confidence?	- Study whether prediction confidence accurately reflects the probability of being correct. - Evaluate calibration techniques under normal conditions and distribution shifts.
24	Knowledge Distillation: How Small Can a Neural Network Become?	- Train a smaller student model to learn from a larger and more powerful teacher model. - Investigate the trade-off between model size, computational efficiency, and prediction performance.
25	Efficient Deep Learning through Quantization, Pruning, and Distillation	- Apply quantization, pruning, and knowledge distillation to reduce the memory and computational requirements of deep neural networks. - Compare these techniques and analyze the trade-off between model efficiency and prediction performance.
26	Continual Image Classification without Catastrophic Forgetting	- Train an image-classification model to learn new classes or tasks over time while preserving previously learned knowledge. - Compare simple sequential training with methods designed to reduce forgetting of earlier classes or tasks.
27	Federated Image Classification under Non-IID Data	- Train an image-classification model collaboratively across multiple simulated clients without directly sharing their local training data. - Investigate how differences in data distributions across clients affect federated-learning performance.

28	Multi-Label Chest X-Ray Classification with Confidence Calibration	- Develop a deep-learning model that can identify multiple findings that may appear simultaneously in a chest X-ray image. - Investigate the effects of class imbalance, prediction thresholds, and confidence reliability on classification performance.
29	Skin-Lesion Classification under Severe Class Imbalance	- Develop an image-classification model for skin lesions when some categories contain far fewer training samples than others. - Compare different sampling, loss-function, and data-augmentation strategies for handling class imbalance.
30	Explainable Retinal Image Classification	- Classify retinal images using deep-learning models and visualize the image regions that contribute most to model predictions. - Compare different explanation methods and analyze whether the highlighted regions are consistent for correct and incorrect predictions.
31	Medical Image Segmentation with Limited Training Data	- Develop deep-learning models for segmenting anatomical structures or abnormalities when only a limited amount of annotated medical images is available. - Compare conventional segmentation models with pretrained models and investigate how training-data size affects segmentation performance.
32	Robust Histopathology Image Classification under Staining Variations	- Classify histopathology images when differences in staining and image appearance are present. - Investigate how color normalization, data augmentation, and pretrained vision models affect classification robustness.
33	ECG Arrhythmia Classification with Deep Sequence Models	- Classify different types of heart rhythms from ECG signals using deep-learning models. - Compare convolutional, recurrent, and Transformer-based approaches and analyze their performance under different evaluation settings.
34	Robust Speech Emotion Recognition with Pretrained Audio Models	- Recognize emotions from speech using conventional audio features and pretrained deep-learning audio models. - Evaluate how noise and differences between speakers affect emotion-recognition performance.
35	Tiny Keyword Spotting under Noise and Resource Constraints	- Recognize short spoken commands using compact deep-learning models suitable for resource-limited devices. - Investigate the trade-off between recognition accuracy, noise robustness, model size, and inference speed.
36	Environmental Sound Recognition under Unseen Acoustic Conditions	- Classify environmental sounds such as sirens, engines, animal sounds, and machinery using deep-learning models.

		- Investigate how well the models perform when test audio contains noise, reverberation, or acoustic conditions not encountered during training.
37	Human Activity Recognition with Multi-Sensor Deep Learning	- Recognize human activities from wearable-sensor signals, such as accelerometer and gyroscope data, using deep-learning models. - Compare different sensor combinations and deep sequence models to determine how multiple sensors affect recognition performance.
38	Deep Time-Series Anomaly Detection under Changing Conditions	- Detect unusual or abnormal patterns in time-series data using deep-learning models. - Investigate how noise, changes in operating conditions, and different detection thresholds affect anomaly-detection performance.
39	Few-Shot Video Action Recognition with Pretrained Video Models	- Recognize human actions in short video clips when only a limited number of labeled training examples are available. - Investigate how pretrained video models, frame-sampling strategies, and training-data size affect action-recognition performance.